Molecular Dynamics Simulations of Bacterial and Plant-Derived Nanocelluloses: Impact on the Properties of Polyhydroxybutyrate Biopolymer

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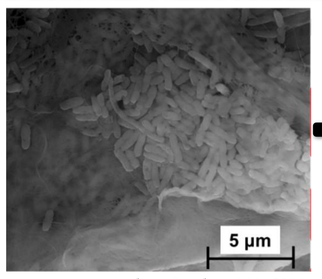
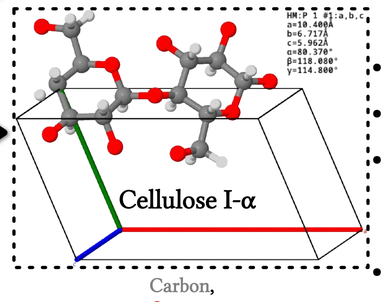
Biopolymers used as **reinforcing materials** for the enhancement of polymeric **matrices** properties → **Homogenization** of **Nanocellulose** in **PHB** Matrices ← **Enhanced biodegradable** materials for **packaging** applications

Background

Bacterial Nanocellulose



Author: Gamboa, MA. 2024.

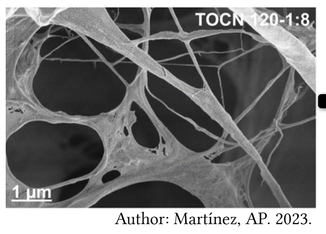


Carbon, Oxygen

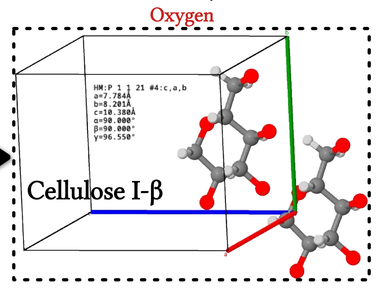
Composed mainly by **Cellulose I-α**

- *High purity*
- *Fine fiber network*
- *Superior mechanical properties*
- *Crystalli. > 60-70%*

Palm Cellulose NF



Author: Martínez, AP. 2023.

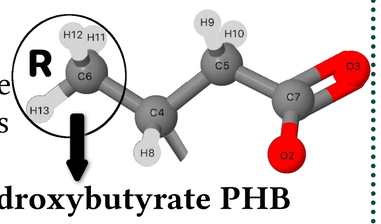


Composed mainly by **Cellulose I-β**

- *Coarser fiber network*
- *Thermal stability*
- *Cost-effective source*

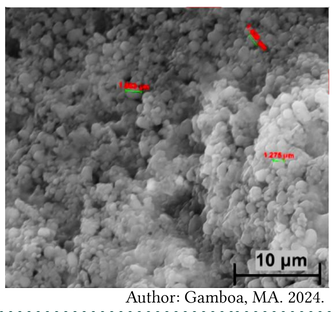
Polyhydroxyalkanoates

General structure of PHA's



Polyhydroxybutyrate PHB

Thermoplastic, Crystallinity>70-80 %



Author: Gamboa, MA. 2024.

SEM Image of PHB

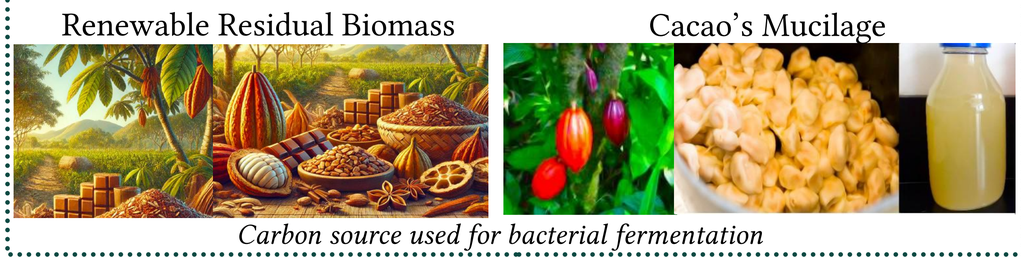
Nanocomposites based on Nanocellulose + PHB Biofilms

Bacterial Nanocellulose — *PHB Films*

Author: Gamboa, MA. 2024. Author: Saavedra, O. 2021.

Suitable for biodegradable packages and other eco-friendly materials.

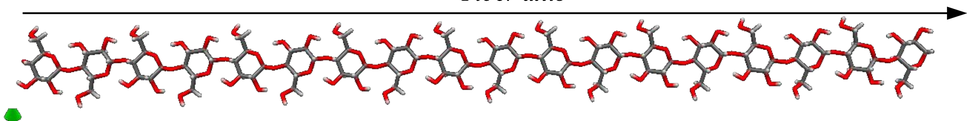
Renewable Residual Biomass — Cacao's Mucilage



Carbon source used for bacterial fermentation

Cellulose and PHB models

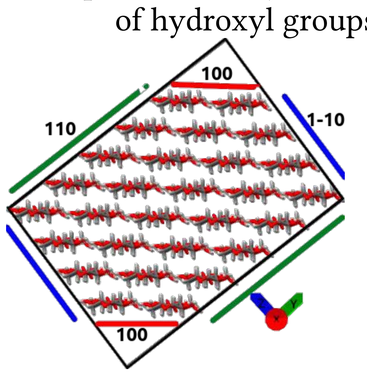
Fiber axis



Cellulose I-α chains with PD = 18 anhydro-glucose units

COD	Formula	Space group	Cell Parameters a, b, c	α,β,γ
4114382	$C_{12}H_{24}O_{10}$	P1	10.4 ± 0.01 Å	80.37 ± 0.05°
			6.717 ± 0.006 Å	118.08 ± 0.05°
			5.962 ± 0.007 Å	114.80 ± 0.05°

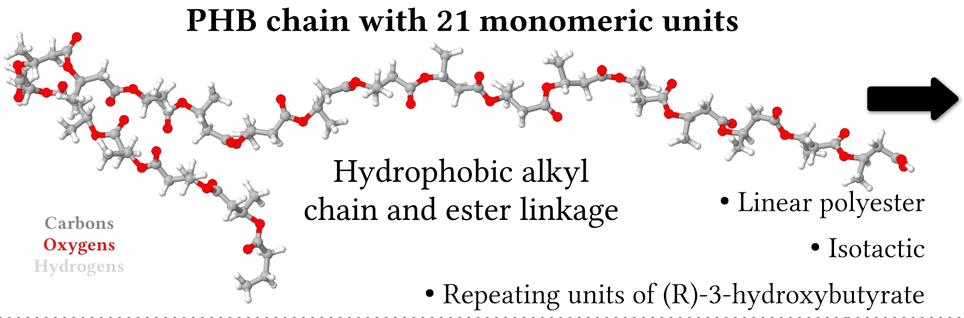
Exposition of crystal face with main number of hydroxyl groups to PHB matrix



(100) Hydrophobic surfaces

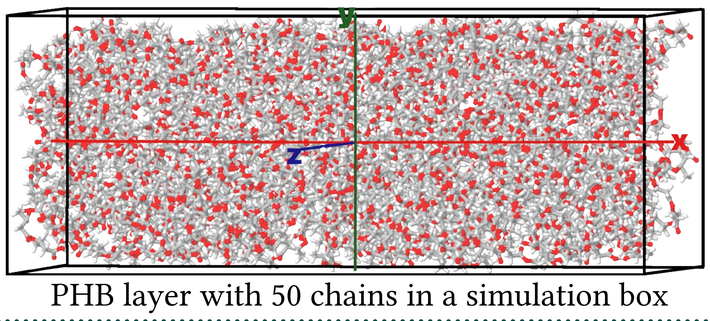
(110) and **(1-10)** Hydrophilic surfaces

PHB chain with 21 monomeric units



Hydrophobic alkyl chain and ester linkage

- Linear polyester
- Isotactic
- Repeating units of (R)-3-hydroxybutyrate



PHB layer with 50 chains in a simulation box

Computational Chemistry Tools

Visualization and Manipulation Tools

PACKMOL, VMD, CHARMM-GUI Input Generator, COD

Simulation software:

GROMACS

Equilibration: NVT ensemble, followed by NPT at 298.15 K

Annealing Process: temperature changes from 303 K to 700 K. at the NPT ensemble.

Cellulose Builder was used for creation of cellulose crystal structures.[4]

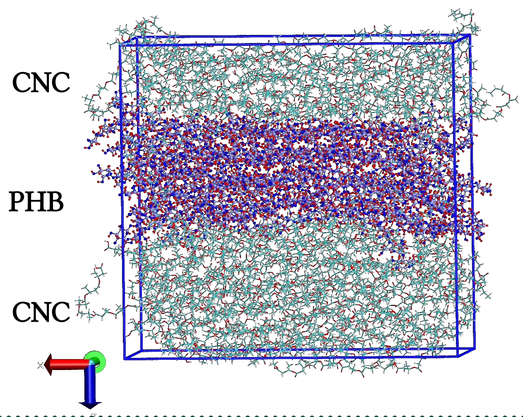
High Performance Computing:

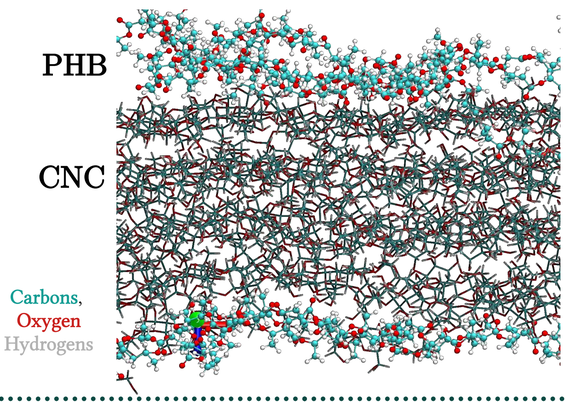
Cluster GUANE — Super Computación y Cálculo Científico UIS

Cluster Rackable SGI — INKARI, UNSA, IAAPP

Design of Cellulose-PHB Interface with -OH groups exposed to PHB

Snapshots of systems after 50 ns of simulation





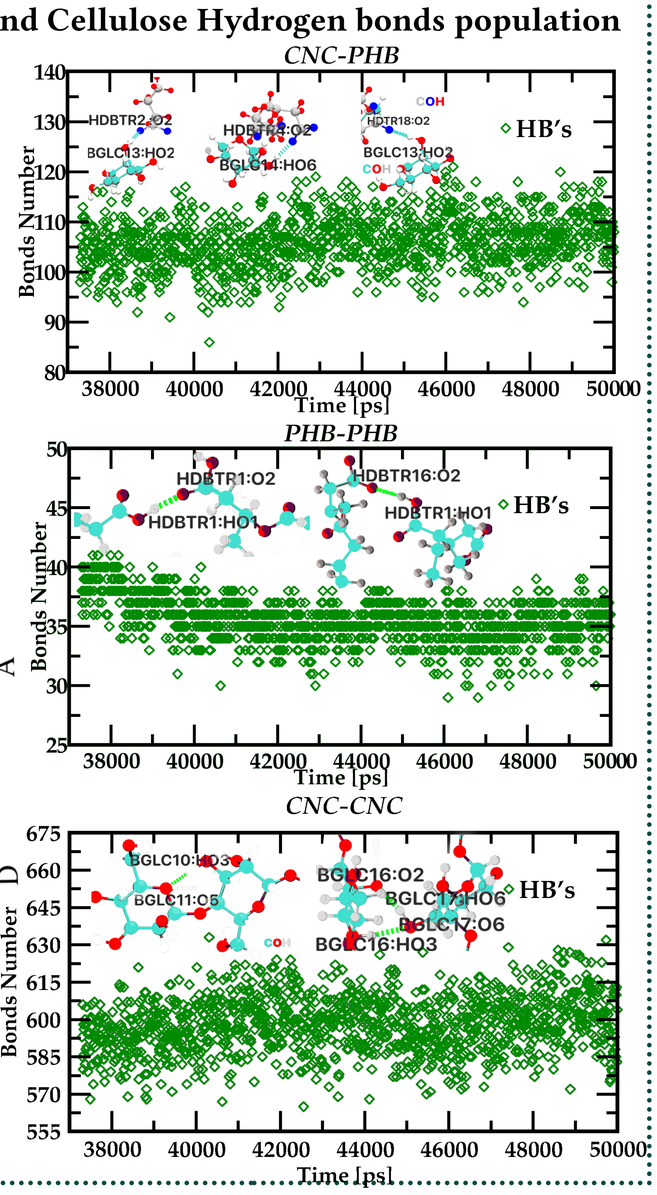
Carbons, Oxygen, Hydrogens

NPT ensemble and time-step of 1 fs. V-rescale thermostat for temperature coupling. c-rescale barostat for pressure coupling.

The *CHARMM36-ff* for *Carbohydrates* was used for the simulations

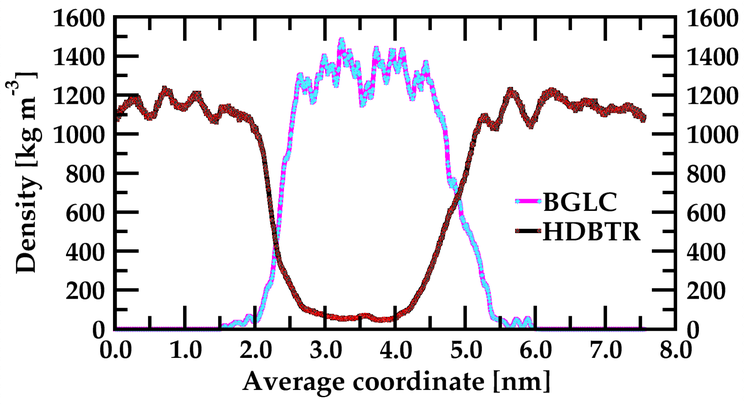
System composition:
18 CNC chains
50 PHB chains
16456 atoms
393.207 nm^3

PHB and Cellulose Hydrogen bonds population



Cut-off for HB's: $r \le r_{HB} = 0.35$ nm, $\alpha \le \alpha_{HB} = 30°$

Density Profiles along interface direction Cell-PHB



Mean Square Displacement MSD and Diffusion Coefficients

$MSD_Ave_{[Cell]}$ = 0.00698 nm²

$MSD_Ave_{[PHB]}$ = 0.0383 nm²

$Diff_Cfte_{[Cell]}$ = 5.594e-07 nm²/ps

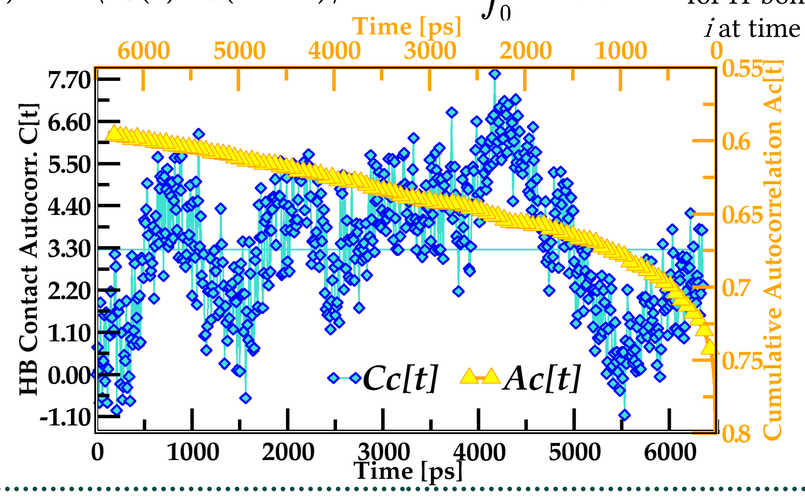
$Diff_Cfte_{[PHB]}$ = 3.954e_06 nm²/ps

Hydrogen Bonds Autocorrelation PHB-Cellulose

Stability and lifetime of hydrogen bonds over time, plays a role in the mechanical stability of the composite material.

- *HB's are more transient, breaking and reforming frequently.*

$$C(\tau) = \langle s_i(t)\, s_i(t+\tau)\rangle \quad \tau_{HB} = \int_0^\infty C(\tau)d\tau \qquad s_i(t) = \{0,1\} \text{ for H-bond } i \text{ at time } t$$



References:

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On-going work:

Simulations are currently running for systems with different numbers of hydroxyl groups exposed at the cellulose surface. More analyses have been performed.

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